



Data Visualization Intro

DSCI 454
Week 1, Lecture 2

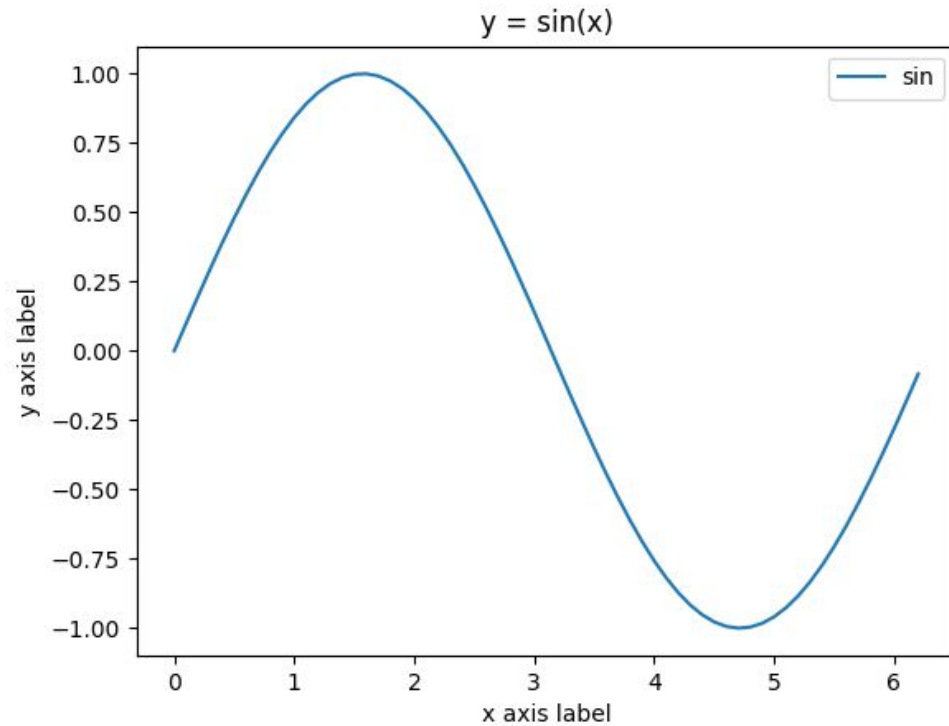
Why Visualize?



- What is this data?

```
[ 0.      0.09983342 0.19866933 0.29552021 0.38941834 0.47942554
 0.56464247 0.64421769 0.71735609 0.78332691 0.84147098 0.89120736
 0.93203909 0.96355819 0.98544973 0.99749499 0.9995736  0.99166481
 0.97384763 0.94630009 0.90929743 0.86320937 0.8084964  0.74570521
 0.67546318 0.59847214 0.51550137 0.42737988 0.33498815 0.23924933
 0.14112001 0.04158066 -0.05837414 -0.15774569 -0.2555411  -0.35078323
 -0.44252044 -0.52983614 -0.61185789 -0.68776616 -0.7568025  -0.81827711
 -0.87157577 -0.91616594 -0.95160207 -0.97753012 -0.993691  -0.99992326
 -0.99616461 -0.98245261 -0.95892427 -0.92581468 -0.88345466 -0.83226744
 -0.77276449 -0.70554033 -0.63126664 -0.55068554 -0.46460218 -0.37387666
 -0.2794155  -0.1821625  -0.0830894 ]
```

Same Question





Why data visualization matters in analytics and policy

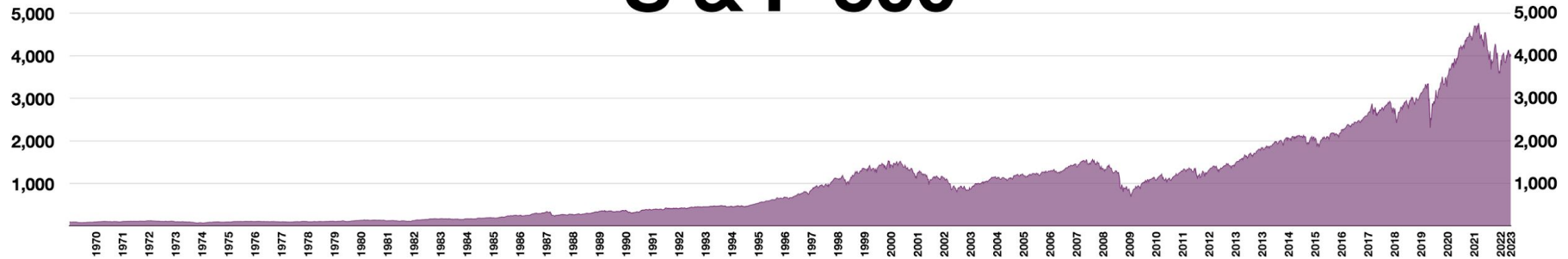
Why Data visualization matters in analytics



- Reveals:
 - Relationships between features, characteristics, attributes, measurements, observations
 - Trends
 - Is performance improving quarter to quarter?
 - Outliers
 - Is this piece of data truly representative?
 - Anomalies
 - Did an error occur?
 - Sometimes easy to miss in logs and big data
 - Clusters
 - Sub-species?
 - Customer types / groups?
 - Often hard to see with just raw table of numbers



S & P 500

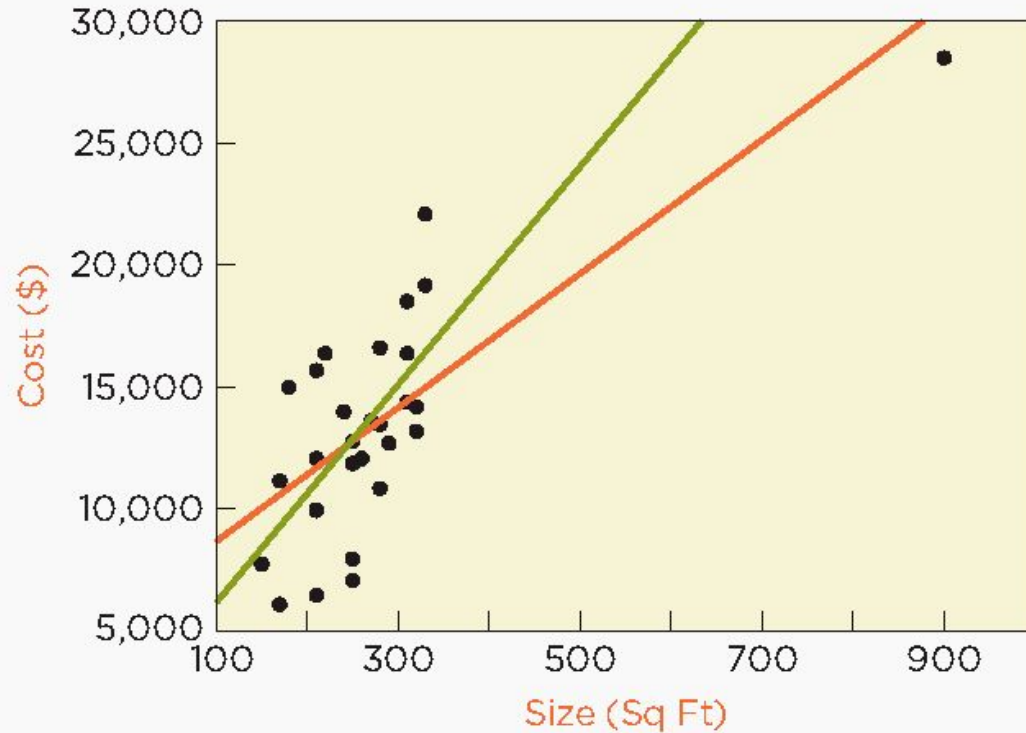


<https://www.investing.com/indices/us-spx-500-historical-data>, CC0,
<https://commons.wikimedia.org/w/index.php?curid=129407392>

Outliers



- Outliers potentially skew your data analysis. For example:
 - A contractor is bidding on a project to construct an 875 square foot addition to a home. How much do they bid?
- The addition of the point at approx. (900, 28,000) changes the trendline to something that no longer fits the rest of the data as well.



Find the Outlier by Age on Titanic

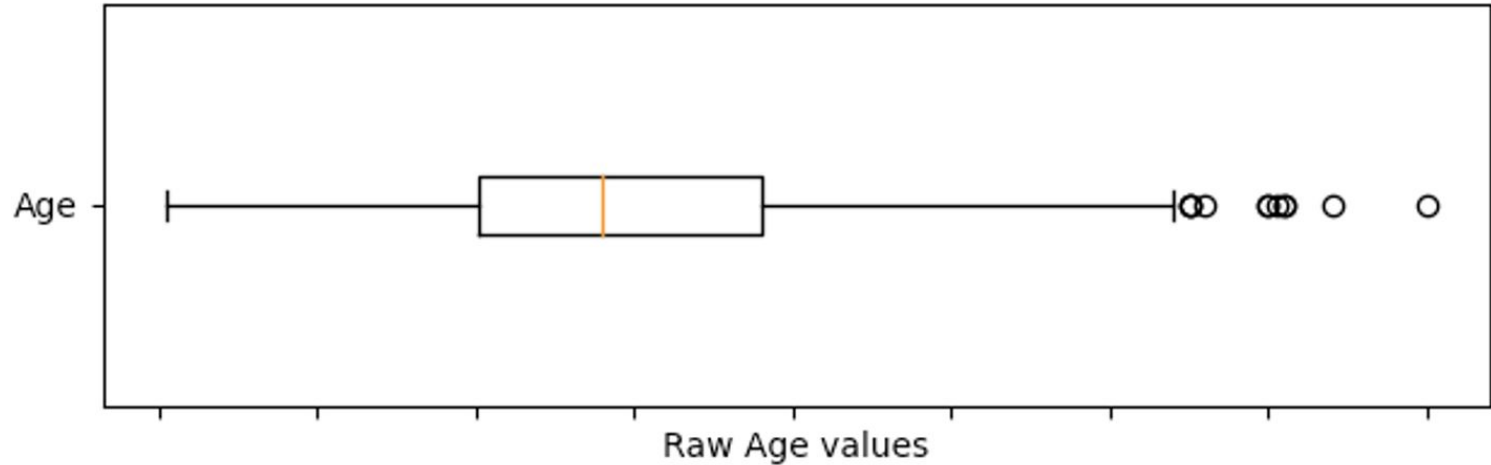


	A	B	C	D	E	F	G	H	I	J	K	L
1	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
2	1	0	3	Braund, Mr. C	male	22	1	0	A/5 21171	7.25		S
3	2	1	1	Cummings, Mr.	female	38	1	0	PC 17599	71.2833	C85	C
4	3	1	3	Heikkinen, M	female	26	0	0	STON/O2. 31	7.925		S
5	4	1	1	Futrelle, Mrs	female	35	1	0	113803	53.1	C123	S
6	5	0	3	Allen, Mr. Wil	male	35	0	0	373450	8.05		S
7	6	0	3	Moran, Mr. J	male		0	0	330877	8.4583		Q
8	7	0	1	McCarthy, M	male	54	0	0	17463	51.8625	E46	S
9	8	0	3	Palsson, Mas	male	2	3	1	349909	21.075		S
10	9	1	3	Johnson, Mrs	female	27	0	2	347742	11.1333		S
11	10	1	2	Nasser, Mrs.	female	14	1	0	237736	30.0708		C
12	11	1	3	Sandstrom, M	female	4	1	1	PP 9549	16.7	G6	S
13	12	1	1	Bonnell, Miss	female	58	0	0	113783	26.55	C103	S
14	13	0	3	Saunderscock	male	20	0	0	A/5. 2151	8.05		S
15	14	0	3	Andersson, M	male	39	1	5	347082	31.275		S
16	15	0	3	Vestrom, Mis	female	14	0	0	350406	7.8542		S
17	16	1	2	Hewlett, Mrs	female	55	0	0	248706	16		S
18	17	0	3	Rice, Master	male	2	4	1	382652	29.125		Q
19	18	1	2	Williams, Mr.	male		0	0	244373	13		S
20	19	0	3	Vander Plank	female	31	1	0	345763	18		S
21	20	1	3	Masselmani,	female		0	0	2649	7.225		C
22	21	0	2	Fynney, Mr. J	male	35	0	0	239865	26		S
23	22	1	2	Beesley, Mr.	male	34	0	0	248698	13	D56	S
24	23	1	3	McGowan, M	female	15	0	0	330923	8.0292		Q
25	24	1	1	Sloper, Mr. W	male	28	0	0	113788	35.5	A6	S
26	25	0	3	Palsson, Miss	female	8	3	1	349909	21.075		S
27	26	1	3	Asplund, Mrs	female	38	1	5	347077	31.3875		S
28	27	0	3	Emir, Mr. Far	male		0	0	2631	7.225		C
29	28	0	1	Fortune, Mr.	male	19	3	2	19950	263	C23 C25 C27	S
30	29	1	3	O'Dwyer, Miss	female		0	0	330959	7.8792		Q
31	30	0	3	Todoroff, Mr.	male		0	0	349216	7.8958		S
32	31	0	1	Urchurtu, D	male	40	0	0	PC 17601	27.7208		C
33	32	1	1	Spencer, Mrs	female		1	0	PC 17569	146.5208	B78	C
34	33	1	3	Glynn, Miss. I	female		0	0	335677	7.75		Q
35	34	0	2	Wheadon, M	male	66	0	0	C.A. 24579	10.5		S

Find the Outlier by Age on Titanic



Titanic dataset: identifying and removing Age outliers



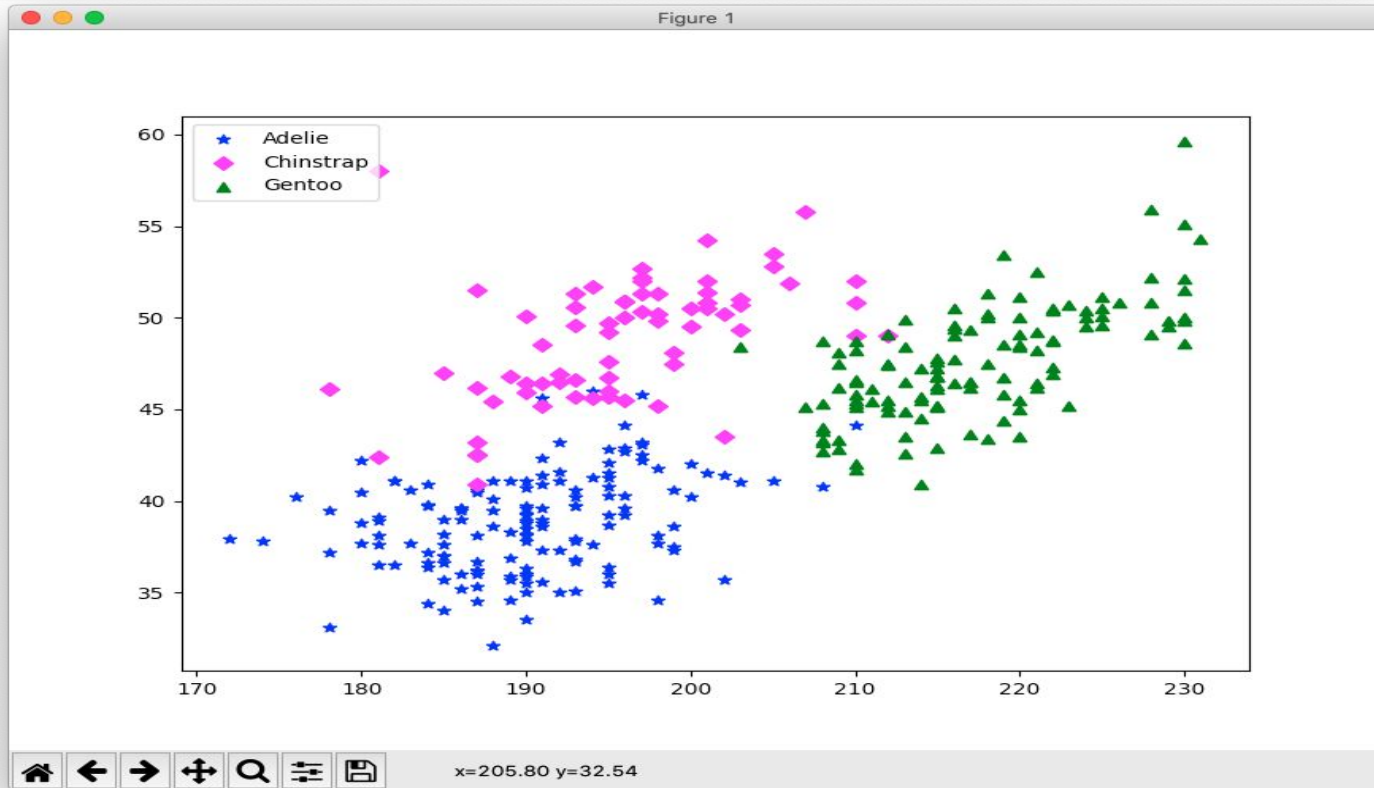
See the clusters of penguin sub-species?



	C	D	E	F	G
1	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex
2	39.1	18.7	181	3750	MALE
3	39.5	17.4	186	3800	FEMALE
4	40.3	18	195	3250	FEMALE
5					
6	36.7	19.3	193	3450	FEMALE
7	39.3	20.6	190	3650	MALE
8	38.9	17.8	181	3625	FEMALE
9	39.2	19.6	195	4675	MALE
10	34.1	18.1	193	3475	
11	42	20.2	190	4250	
12	37.8	17.1	186	3300	
13	37.8	17.3	180	3700	
14	41.1	17.6	182	3200	FEMALE
15	38.6	21.2	191	3800	MALE

◀ ▶ penguins +

See the clusters of penguin sub-species?

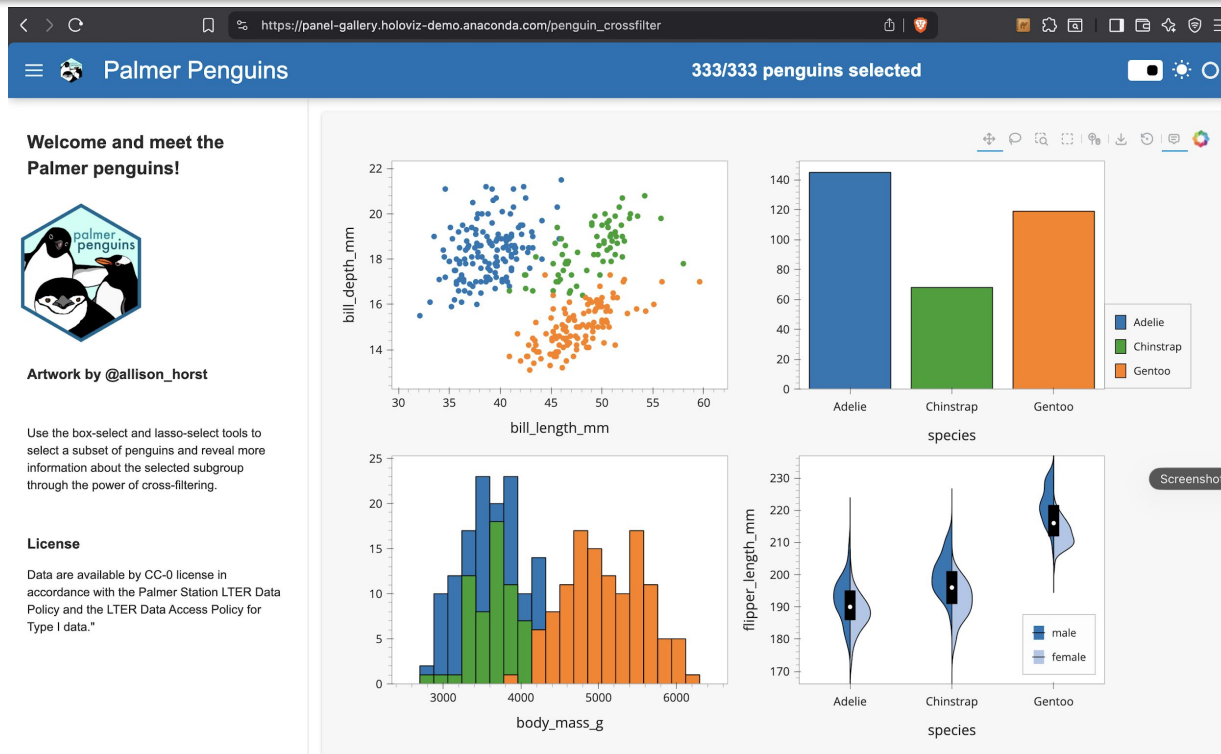


Why Data visualization matters in policy



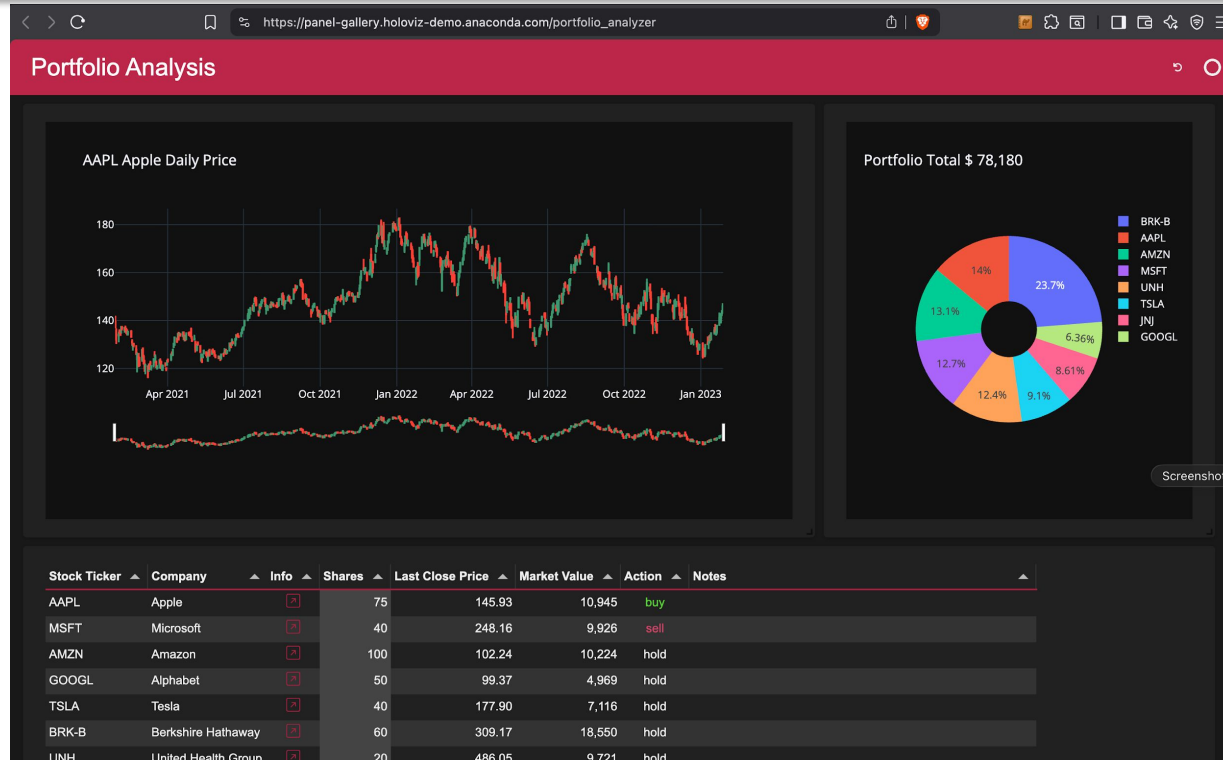
- Data-Driven Decisions
 - Justified, fair, normalized, standardized, objective, evidence-based, quantified
 - rubrics, job classification guides, law-making, etc.
- Trust
 - Confidence and accountability when everyone, incl. the public can see the data
- Dashboards
 - Immediate, on-going, actionable information
 - Automated, mechanized information radiators
 - Different metrics and views for different audiences / consumers
- Technical information for non-technical people
 - Democratization of data

Example Dashboard 1



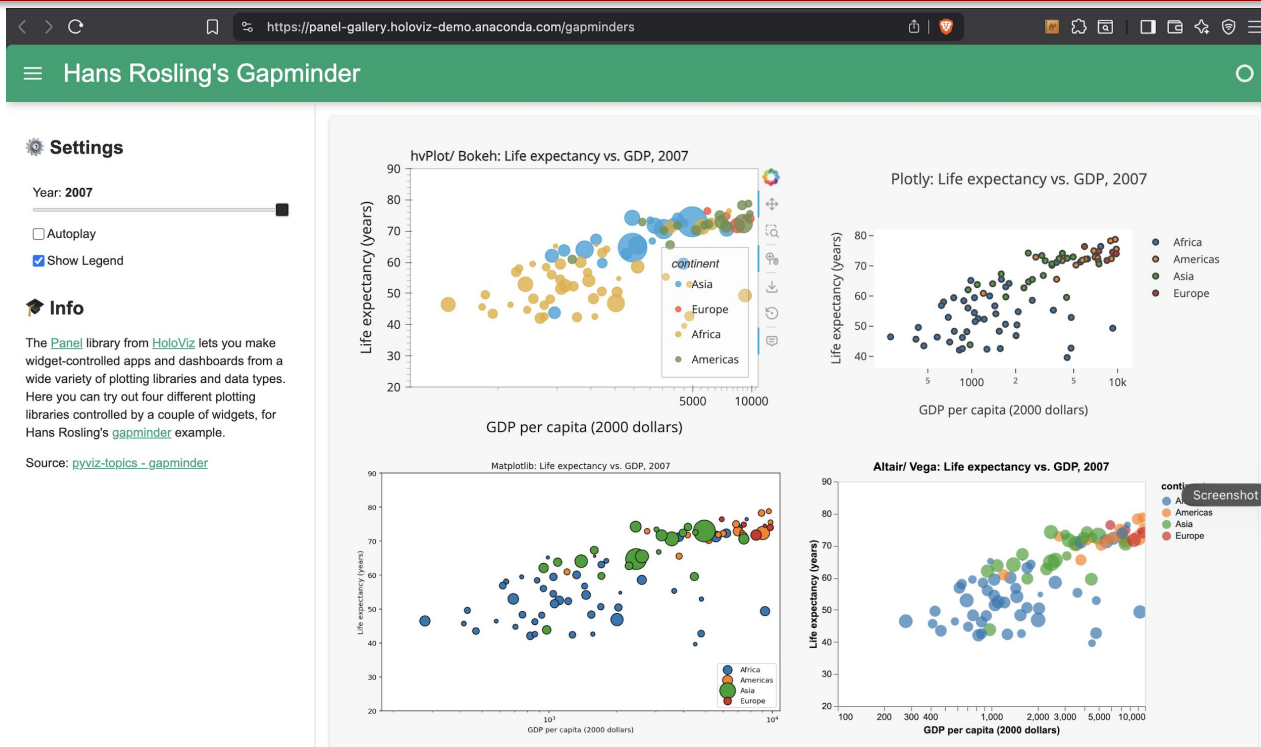
- https://panel-gallery.holoviz-demo.anaconda.com/penguin_crossfilter

Example Dashboard 2



- https://panel-gallery.holoviz-demo.anaconda.com/portfolio_analyzer

Example Dashboard 3



- <https://panel-gallery.holoviz-demo.anaconda.com/gapminders>



How visualization supports reasoning where code, ML, or AI fall short

Raw Data: Tips



	total_bill	tip	sex	smoker	day	time	size	sex_cat	smoker_cat	day_cat	time_cat
0	16.99	1.01	Female	No	Sun	Dinner	2	1	1	3	1
1	10.34	1.66	Male	No	Sun	Dinner	3	0	1	3	1
2	21.01	3.50	Male	No	Sun	Dinner	3	0	1	3	1
3	23.68	3.31	Male	No	Sun	Dinner	2	0	1	3	1
4	24.59	3.61	Female	No	Sun	Dinner	4	1	1	3	1
..	...	...	...	...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3	0	1	2	1
240	27.18	2.00	Female	Yes	Sat	Dinner	2	1	0	2	1
241	22.67	2.00	Male	Yes	Sat	Dinner	2	0	0	2	1
242	17.82	1.75	Male	No	Sat	Dinner	2	0	1	2	1
243	18.78	3.00	Female	No	Thur	Dinner	2	1	1	0	1

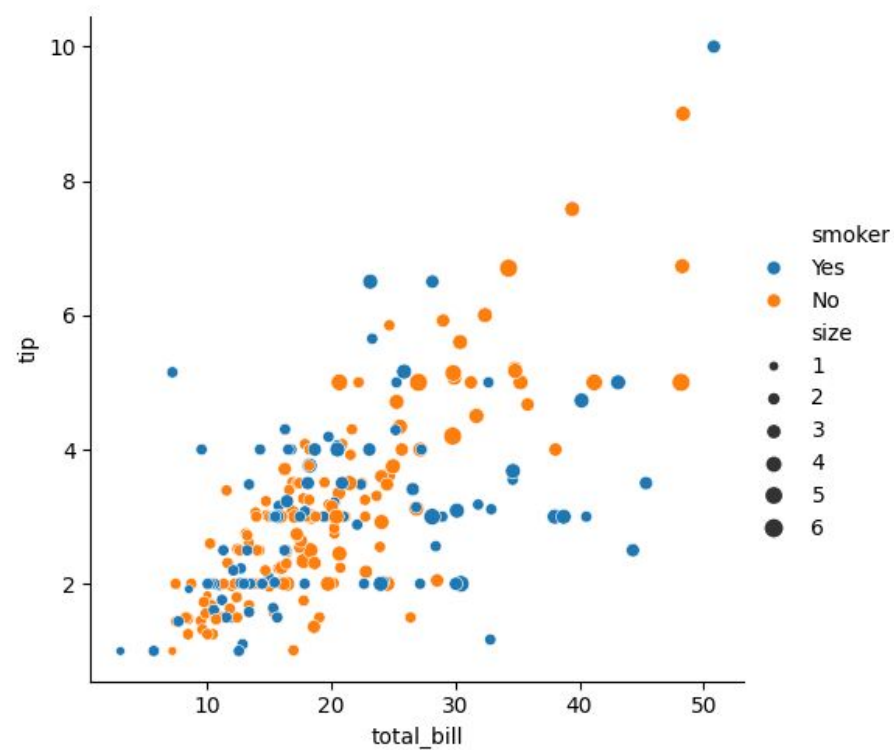
[244 rows x 11 columns]

- Can you see trends, relationships, and correlations of the various features/attributes to target/output/label/prediction at a glance?



Statistical relationships at a glance: Scatter

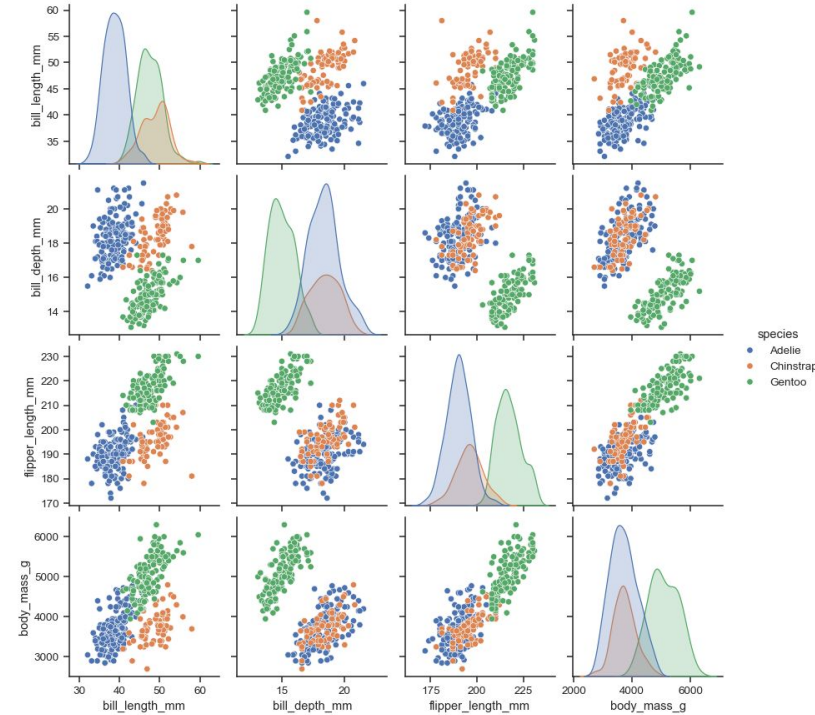
- Here is a seaborn scatterplot of tip amount vs total bill amount where the color is smoker vs non-smoker and the size of the marker is the party size
- <https://seaborn.pydata.org/tutorial/relational.html>





Statistical relationships at a glance: Pairplot

- Pairplots plot “every feature vs every feature” or “every feature vs target” for identifying distributions, trends, relationships, and driving features (correlations, etc.) at a glance
- <https://seaborn.pydata.org/generated/seaborn.pairplot.html>

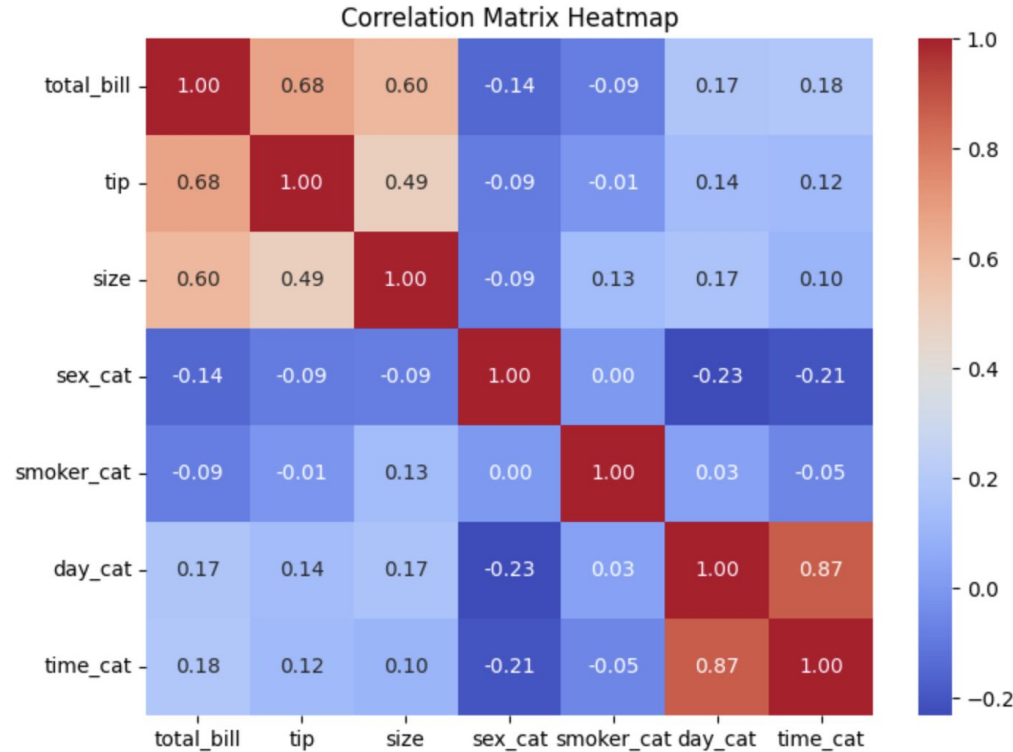


Correlations at a glance: Heatmaps



Feature-to-Target Label Correlation Table:

tip	1.000000
total_bill	0.675734
size	0.489299
day_cat	0.135499
time_cat	0.121629
smoker_cat	-0.005929
sex_cat	-0.088862





Data-driven vs intuition-based decisions

Data-driven decisions



- **What:** Objective information
 - patterns, trends, outlier and anomaly detection
 - metrics guide and iterate on choices
- **Pros:** Repeatable, evidence-based so minimizes cognitive bias, iterative progress measurement, optimization
- **Cons:** Can miss human factors or emerging trends w/o historical data
- **Usage:** Heavily invested / high stakes, new or emerging markets, measuring performance (KPIs), optimizing and iterating on existing processes and previous decisions

[-https://www.ibm.com/think/topics/data-driven-decision-making](https://www.ibm.com/think/topics/data-driven-decision-making)



Intuition-based decisions

- **What:** Decisions based on:
 - "gut feelings"
 - "instinct"
 - experience
- **Pros:** Fast, flexible, utilizes existing expertise
- **Cons:** Not necessarily evidence-based so prone to cognitive biases, human emotions, and incomplete information.
- **Usage:** Quick answers needed, new products or ideas, or non-existent or insufficient data



Real-world examples: data-driven vs intuition-based decisions

Real world examples of data-driven decisions



- Google
 - “People Operations” (recruiting, retaining) -> “People Analytics” using data metrics and visualizations (*Storytelling w/ Data*, Knaflitz, 2015, p. ix)
- Video Streaming Services / Online retailers
 - Use viewing / purchase history to personalize recommendations
- Brick and mortar retailers
 - Perform sales data analysis to optimize inventory
 - <https://corporate.walmart.com/news/2025/10/29/walmart-data-ventures-shapes-the-next-era-of-insight-driven-retail-innovation>
- New York Fire Department (FDNY)
 - Use data on past fires to score and prioritize high-risk buildings for inspection
 - <https://www.govtech.com/public-safety/new-york-city-fights-fire-with-data.html>
- Credit Card Companies / Investment firms
 - Employ predictive analytics to spot fraud / risky investments early

Real world examples of intuition-based decisions



- Apple
 - Steve Jobs relied on “gut feeling” for early product design (e.g. original iPhone or original iTunes) when no historical market data existed -Steve Jobs Biography
 - <https://www.smithsonianmag.com/arts-culture/how-steve-jobs-love-of-simplicity-fueled-a-design-revolution-23868877>
- Amazon
 - Bezos’ ultimate decision to greenlight Prime was “based on intuition” despite negative financial model data
 - <https://www.cnbc.com/2018/11/19/jeff-bezos-simple-strategy-for-answering-amazons-hardest-questions-.html>
- Starbucks
 - Schultz’s vision for Italian-style coffee cafés in America was not supported by data at the time
 - <https://www.britannica.com/money/Starbucks#ref330440>

Combining intuition and data



- Sometimes the best decisions are made with a combination of both intuition and data
 1. Make an intuition-based decision in the absence of historical data
 2. Then use data-driven analytics, metrics, and dashboards from the previous slides to guide, iterate, and optimize going forward
- This course will focus on using visualizations to aid in making data-driven decisions

IBM's Data-Driven Decision-Making Steps



1. Define objectives
 2. Identify, prepare and collect data
 3. Organize and explore:
 - **“Visualizing the data helps identify patterns, outliers and trends that are not immediately obvious from raw data.”**
 4. Perform data analysis
 5. Draw conclusions
 6. Implement and evaluate
- <https://www.ibm.com/think/topics/data-driven-decision-making>



1. Define Objectives

- The 2 A's: Audience and Ask
 - (*Storytelling w/ Data*, Knafllic, 2015, p. 21)
- Who is your Audience?
 - Data Scientists?
 - Executives?
 - Customers?
 - What is your relationship to them?
- What is your Ask?
 - What do you need your audience to know or do?
 - Adopt a tool?
 - Pivot to a new market?
 - Increase price?
 - Try a new product?
 - Could be a recommendation or just a discussion
- Tailor your presentation/visualization to your Audience and Ask



2. Identify, prepare, and collect data

- In the wild (industry and higher academia), the starting point is rarely a curated dataset from Kaggle or similar
- Data will need to be mined
 - Internal or external
 - Internal employee data, survey data
 - External APIs from service providers or screen scrapers
 - (more on both later this semester)
- Multiple sources may need to be aggregated into one in a relational SQL or NoSQL database, etc.
- Authentication and Authorization
- EULAs, Terms of Service, Throttling, etc.



3. Organize and explore

- Records could be messy in a number of ways:
 - **Duplicate** records could exist.
 - Values could have been entered incorrectly - i.e. **malformed** - on some records.
 - Values could be **missing** from records.
 - There could be records which contain values far away from the rest of the dataset; **outliers**.
 - ITP/TAC-216: Applied Python
- Exploratory vs Explanatory Data Analysis (more later)
- **Visualize, Visualizations -> Presentations**
 - Raw Data -> Information
 - Note: Multiple visualizations may get created and never used in final presentation



4. Perform Data Analysis

- See previous slides:
 - Correlations,
 - trends,
 - outliers,
 - anomalies
- Combined w/ Step 2 and 3 to:
 - explore,
 - filter,
 - aggregate (sum, product, average, mode, median, min, max, etc.),
 - Curate, tailor to Audience and Ask
- More later in the semester



5. Draw Conclusions

- Based on the evidence from the previous steps, now ready to provide actionable, insightful, data-backed recommendations
- Data is now information from the data analysis in Step 4
- Visualizations from Steps 3 and 4 are now aids to convey this analysis information



6. Implement and Evaluate

- Action Plans developed and implemented
 - e.g. Marketing campaign targeted to an overlooked customer type identified by Step 4's data analysis
- S.M.A.R.T. (Specific, Measurable, Attainable, Realistic, Timely) goals and metrics (Key Performance Indicators (KPIs)) are created and continuously tracked and improved via Dashboards and feedback loops
 - **Specific**: Instead of “increase sales”, “increase sales by 15%)
 - **Measurable**: which metrics/KPIs will show this?
 - **Attainable/Realistic**: can we actually do this over this timeframe?
 - **Timely**: Timeframe: e.g. “before the end of Q4”
 - e.g. “**Increase** sales of Product X by **15%** by the end of **Q4** as shown by the following **KPIs** (x,y,z) **visualized** on this **Dashboard** ABC” which is continuously **updating** radiating **information** in a feedback loop
- **Iterate, Iterate, Iterate**

Questions?

